Introduction

I. Introduction to Systems

1. Analog vs. Digital Systems

- **Analog System:** Deals with analog quantities.
  - *Characteristics:* Continuous nature.
  - *Examples:* Temperature, Pressure, Flow, Speed, Mass.
- **Digital System:** Deals with physical quantities in a modern context.
  - *Characteristics:* Discrete steps (e.g., a valve changing in specific steps rather than continuously).

    ### 2. Why do we study Analog and Digital Systems?
- To understand the transition from continuous natural phenomena to discrete computing processing.

II. Core Terminology

- **Logic:** A set of principles or rules used for reasoning.
- **Design:** A plan or strategy for the creation of products or services.
- **Digital Circuit:** A specific type of design that implements logic rules using electronic components.

III. Course Objective: What do we do in DLD?

Primary Goal: Design digital circuits using **Boolean Logic**.
How to design digital systems:

1. **Machine Language:** Electronic machinery fundamentally understands only `0` and `1` (Binary).
2. **Representation:** While machines use binary (`0`, `1`), humans often use **Hexadecimal** to make large binary numbers easier to understand and read.
3. **Foundational Knowledge:** To master DLD, one must understand:
   - Binary Systems
   - Octal Systems
   - Number Systems in general

IV. Circuit Classifications

Digital logic is divided into two primary types of circuits:

1. **Combinational Circuits:** Output depends only on the current input.
2. **Sequential Circuits:** Output depends on current input and past history (memory).